

ED212-Solid State Devices

Course Credit: 3-0-2-4

Course Objective: This course introduces review of semiconductor devices, basic properties of semiconductors and their conduction process

Course content:

Module 1: Introduction to Semiconductors Physics

Module 2: PN Junctions

Module 3: Semiconductor devices, Photodiodes, Zener diodes

Module 4: Applications of semiconductor devices

Module 5: Solar cell

Module 6: BJT and MOSFET characteristics

Module 7: LED and Laser and compound semiconductors devices

Books:

1. S. M. Sze - Semiconductor Devices: Physics & Technology (2nd Edition, Wiley, 2002).
2. Greg Parker (2004) - Introductory Semiconductor Device Physics. IOP.

Assessment method: Two in-semester examinations, a final examination, lab assignments and project.

Evaluation scheme: (i) In-Sem Exam: 15%, (ii) In-Sem Exam: 20%, (iii) End-Sem Exam: 40%, and (iv) Labs: 25% .

Course Outcomes:

1. The students shall be able to understand the basic fundamentals of semiconductors physics.
2. The students shall be able to comprehend different semiconductor devices.
3. The students shall be able to use semiconductor devices for different applications. This shall facilitate students to make state-of-the-art practical research projects and meet industrial demands.

P1	P2	P3	P4	P5	P6	P7	P8	P9	P10	P11	P12
X	X	X	X	X							X